Prodesy

An LLM-maintained wiki on protein design — book edition

Compiled from the Prodesy wiki

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Introduction — the field at a glance

Evolving synthesis of the Prodesy wiki. Current scope: the deep-learning core (batch 1) plus the generative-design expansion (batch 2). Will broaden as more of `sources/ProteinDesign/` is ingested.

The through-line

A single pipeline now organizes the field:

> **generate a backbone → design its sequence → predict/filter before you build**

- **Generate** — Diffusion generative design: Watson 2023 (fine-tuned RoseTTAFold) and, independently, Ingraham 2023 (bespoke polymer-aware diffusion) sample novel, steerable backbones.
- **Sequence** — Sequence design (fixed-backbone): Dauparas 2022 assigns a foldable sequence to a backbone in seconds, displacing Rosetta.
- **Filter** — Structure prediction: Jumper 2021 confidence scores decide, *in silico*, which designs are worth synthesizing.

Bennett 2023 shows the payoff concretely: dropping ProteinMPNN + AF2 filtering into the older Rosetta binder workflow raised success ~10×. Binder design is the clearest application of the whole stack.

How the pipeline was built and where it's going (batch 2)

- **Precursors.** Before diffusion, the same goals were met by **network hallucination** and **inpainting**. Anishchenko 2021 established the method — invert a structure predictor by optimizing a random sequence until the network's own prediction sharpens — and Wang 2022 (Motif scaffolding (functional-site design)) and Wicky 2022 (Symmetric assembly design) are the same engine under two additional loss terms. RFdiffusion later absorbed all of it as conditional-diffusion tasks: what was a bespoke objective becomes a conditioning input.
- **Symmetric assemblies are now a chapter.** Symmetric oligomer design traces the longest continuous arc in the wiki, from Harbury 1993 through the parametric era (Huang 2014, Thomson 2014, Boyken 2016, Courbet 2022) to hallucination and conditional diffusion.
- **All-atom is the current frontier**, pursued by ≥3 distinct routes: atomize-in-predictor (Krishna 2024), intrinsic side-chain **superposition** (Chu 2024, Huang lab), and efficient **token+atom** generation (Butcher 2025, ~10× cheaper, DNA binders). This mirrors, on the design side, AF3's all-atom move on the prediction side.

- **Sequence design is converging on codesign.** Dauparas 2022 and the learned-potential Anand 2022 (Huang lab) both showed learned energies beat Rosetta; Protpardelle now folds sequence *into* generation.
- **Applications reach harder targets.** Vasquez-Torres 2023 extends Binder design to floppy peptide hormones via **partial diffusion**, and turns binders into biosensors.

Cross-cutting observations

- **Prediction ⇄ design convergence.** Prediction models are reused as design oracles (AF2 filtering) and as generators (RoseTTAFold → RFdiffusion). Meanwhile prediction itself adopted diffusion in Abramson 2024. The two problems are merging onto shared machinery.
- **Independent replication of the paradigm.** RFdiffusion (Baker lab) and Chroma (Generate Biomedicines) arrived at diffusion-based design separately — a sign the approach is robust rather than a single group's trick. A worthwhile open comparison: fine-tuned-predictor vs. purpose-built noise prior.
- **Validation discipline is the real unlock.** The recurring lesson across sources is not just better generators but cheap, reliable *in silico* filtering that collapses wet-lab search space.
- **Conditioning moved from the coordinates into the sampling process.** Symmetric design is the cleanest place to watch this happen: symmetry was first *built into* a coordinate generator (Huang 2014 (suppl.)), then *rewarded* in a loss against a predictor (Wicky 2022), then *projected* onto the state at every denoising step (Watson 2023 (suppl.)). The wiki's reading, developed in Symmetric oligomer design, is that this shift traded a geometry bottleneck for an interface-quality bottleneck — coverage improved far faster than yield.
- **The pre-deep-learning literature is doing real work.** Batch 3 showed that the wiki's synthesis was resting on Baker-lab-2022-onward sources alone for a problem with a thirty-year history. Where a chapter's claims trace only to recent papers, that is worth treating as a gap rather than a summary.
- **Generative models inherit priors nobody asked for.** Anishchenko 2021 modelled every design as a monomer and got homo-oligomers and soluble aggregates anyway, because trRosetta had learned the sequence signatures of native protein–protein interfaces and did not disfavour surface hydrophobics enough. The same bias shows up as adversarial surfaces in Wicky 2022. The wiki's reading is that a predictor-derived generator carries its training distribution's interface statistics into every design, which is sometimes the capability and sometimes the bug — see Symmetric oligomer design §5.1.

Open threads to pursue (next batches / queries)

- **Prediction ecosystem** (batch 3): ColabFold, Foldseek, lDDT, Boltz/Boltzgen — the tooling around the predictors used as design oracles.
- **Surface-fingerprint design** (MaSIF, Gainza Nature + Nature Methods) — a distinct, non-diffusion paradigm for interaction design to contrast with Binder design.
- **Antibody-specific design** (Bennett RFdiffusion antibodies +supp, IgLM) as a sub-branch of Binder design.
- **Remaining generative models**: Boltzgen, Protein Hunter, DidiKreis test-time compute — to keep Diffusion generative design current.
- Add a `benchmarks/` thread (CASP, sequence recovery, binder success rates, the RFD3 PPI/DNA/small-molecule/AME suites) to make cross-paper numbers comparable.
- **Protein dynamics** (Kern lab papers already in folder) — a different lens (conformational ensembles) largely absent from the current design-centric synthesis.

Assemblies, Scaffolds and Validation

Symmetric oligomer design

A chapter on the design of protein assemblies in which several copies of one designed chain are arranged under a **point-group symmetry** — cyclic (C*n*) rings, dihedral (D*n*) stacks, and the cubic groups (tetrahedral, octahedral, icosahedral). The interesting thing about symmetric homo-oligomers, from a design standpoint, is that they let one small design decision — a single protomer and a single interface — determine a much larger object. The interesting thing historically is that this problem has been solved three separate times, by three genuinely different mechanisms, in roughly thirty years: symmetry as **geometry** (imposed exactly, by construction), symmetry as an **objective** (rewarded, by optimization against a structure predictor), and symmetry as a **constraint on a sampling process** (enforced at every step of denoising). That progression, and what each step cost and bought, is the argument of this chapter.

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1. Why symmetry is a design primitive

Symmetric assemblies are not a niche of the natural protein world. Cyclic protein oligomers alone constitute nearly **30% of all deposited PDB structures**, and they participate in almost every biological process (Wicky 2022). The reason usually offered is energetic: symmetric interfaces can be favourable under evolutionary pressure, and natural proteins accordingly oligomerize into linear, cyclic, dihedral, cubic, planar and spatial symmetries (Luo 2016).

For a designer the appeal is combinatorial. If *n* copies of one chain are arranged under a point group, then designing one protomer and one protomer–protomer interface specifies the entire assembly; the relative orientation of two adjacent subunits together with the complementarity of their contact interface determines the multiplicity of the whole complex (Luo

2016). A designed ring is therefore cheap in design effort relative to its size — the property that makes symmetric oligomers attractive as vaccine platforms, delivery vehicles and catalysts (Watson 2023), as building blocks for nanocage assemblies and as substrates for small-molecule binding and catalysis (Wicky 2022).

The costs are equally structural. An error in the single designed interface is repeated n times, so failure modes are amplified rather than averaged out. And the geometry is unforgiving: for fusion-based approaches, a specific arrangement between the symmetry axes of two different oligomeric proteins is required for the design to be accurate at all (Luo 2016). Symmetry makes the design problem smaller and the tolerance for error smaller at the same time.

2. The pre-learning era — borrow, parameterize, or dock

Before deep learning there were three ways to obtain a symmetric assembly, and it is worth being precise about them because the later methods are best understood as removing specific constraints that each imposed.

Borrow the symmetry. The oldest computational route takes symmetry from nature rather than creating it. Yeates and co-workers introduced **genetic oligomeric fusion**: two natural oligomeric domains — a dimeric M1 matrix protein and a trimeric bromoperoxidase — held in a predetermined orientation by a semirigid helical linker, self-assembling into a **12-subunit tetrahedral cage** when the geometric rule that the 3-fold and 2-fold rotational axes intersect at an appropriate non-orthogonal angle was satisfied; substituting a dimerizing carboxylesterase for the trimeric partner produced filaments instead (Luo 2016). A later refinement of the same design yielded a homogeneous **12-subunit, 16-nm cage with a 5-nm central opening**, resolved crystallographically, and the approach was extended to porous cubic cages — a KDPGal–FkpA fusion with **225 Å outer and 132 Å inner diameter** under octahedral symmetry — and to symmetry-matching fusions producing 1D filaments, 2D layers and 3D crystals (Luo 2016).

Fusion is powerful precisely because it inherits well-characterized, stable natural interfaces. Its limitation is the same fact: the designer chooses among the symmetries nature already provides, and the conformational flexibility of the borrowed building blocks can divert the assembly. The KDPGal–FkpA cube is the clean illustration — native mass spectrometry found **12-subunit tetrahedral and 18-subunit prism assemblies alongside the intended 24-subunit cube** (Luo 2016).

Parameterize the backbone. The second route generates the symmetric backbone from a closed-form geometric description. Its conceptual prerequisite is the demonstration that oligomeric state is *encodable*: Harbury 1993 systematically mutated the hydrophobic **a** and **d** core positions of the dimeric GCN4-p1 leucine zipper in concert and obtained peptides forming **two-, three- and four-stranded** structures, showing that the geometry of buried apolar side chains — not merely the hydrophobic/polar pattern — selects the number of chains.

Two decades later this became a generative procedure. Huang 2014 built backbones directly from the **Crick coiled-coil equations**, sampling the supercoil twist ω0, helical twist ω1, supercoil radius R0, per-helix phases and axial offsets on a grid, running Rosetta sequence design at each grid point, and filtering with Rosetta *ab initio* prediction before synthesis. The resulting designs — including a **pentameric parallel left-handed five-helix bundle** — were extremely stable (**extrapolated ΔG_fold > 60 kcal/mol**) with crystal structures showing nearly identical core packing to the models. The crucial mechanical detail is in the supplement: helices are arrayed symmetrically about the z axis, so for an n-helix bundle the superhelical phase takes values $\{0, 2\pi/n, \ldots 2(n-1)\pi/n\}$ (Huang 2014 (suppl.)). **Cyclic symmetry is exact by construction** — a property of the coordinate generator, never optimized for and never checked afterwards.

Published alongside it, Thomson 2014 applied a related geometric logic to the sparse region above four helices. Classical coiled coils of two to four helices account for **more than 98% of known coiled-coil structures**; α-helical barrels, in which each helix contacts two neighbours through *two* independent hydrophobic seams, are the remainder, and most of their possible structures had never been observed. Enumerating the ways two seams can be accommodated in a heptad repeat, scoring with a knowledge-based function and building atomistic models produced discrete water-soluble **C5, C6 and C7** assemblies whose crystal structures matched the models — CC-Pent being the first crystal structure of a designed pentameric barrel and CC-Hept the first heptamer. That paper also introduced a theme worth carrying forward: the **channels are chemically defined and their diameters track oligomer state**, so a ring's lumen is itself a designable feature.

Dock and design. The third route specifies a protomer in advance, rigid-body docks copies of it into a higher-order symmetric arrangement, then optimizes the interface to make the assembled state low-energy. This was the mainstream approach, and Wicky 2022 characterizes the state of the art it inherited in exactly these terms: current approaches required the structure of the monomer to be specified in advance and, **with the exception of parametrically designed helical bundles**, involved rigid-body docking of previously characterized monomers followed by interface optimization.

That single sentence names the binding constraint of the whole era. Because the protomer had to be specified first, the accessible space of oligomers was bounded by the space of already-solved monomers — which in particular excluded assemblies with **more intertwined chains**, where protomer and interface cannot be separated (Wicky 2022).

3. Specificity as a second axis

Getting the geometry right is not the same as getting the *addressing* right. In DNA, specificity comes from modular hydrogen bonds in the core of the double helix; in proteins it comes largely from buried hydrophobic packing complemented by irregular peripheral polar interactions (Boyken 2016). A set of designed coiled-coil oligomers with hydrophobic cores may each fold correctly and still cross-assemble promiscuously.

Boyken 2016 added the missing axis with **HBNet**. Buried polar interactions are hard to design for two compounding reasons: the partially covalent hydrogen bond restricts donors and acceptors to narrow ranges of orientation and distance, and nearly all buried polar atoms must find a partner or incur a considerable desolvation penalty. Conventional design algorithms make this worse by expressing system energy as a sum of *pairwise* residue interactions, whereas a hydrogen-bond network is inherently many-body. HBNet instead precomputes hydrogen bonds and traverses the resulting graph to enumerate all networks of **three or more residues**, rejecting any network leaving a buried donor or acceptor unsatisfied. Hosting those networks in parametrically generatable coiled coils — which have repeating geometric cross sections, so a network can be reused at equivalent positions around the ring — produced **dimers, trimers and tetramers** built as two concentric rings of helices, in triangular, square and supercoiled topologies not previously seen. **Ten crystal structures** confirmed the designs, with left-handed trimers matching at sub-ångström RMSD and dimer networks as close as **0.39 Å**, and the networks conferred **interaction specificity in vivo**.

This is the point at which symmetric-oligomer design stops being "can we build a ring" and becomes "can we build a set of rings that do not talk to each other" — the prerequisite for anything multi-component.

4. The ceiling of the hand-built approach

Courbet 2022 is the best available measure of how far the parametric-plus-Rosetta toolchain could be pushed, and it is instructive that the paper is not really about making symmetric oligomers at all — it is about making symmetric oligomers *move*.

The premise, as Courbet 2022 states it, is that designed nanomaterials to that point — the paper names icosahedral nanocages and two-dimensional arrays — are assemblies whose components have **fixed orientations** relative to one another; the goal was a kinematic pair with internal degrees of freedom. The design proceeds hierarchically: rigid components first, directed self-assembly into hetero-oligomeric complexes second, shaping of the multistate energy landscape along the mechanical degrees of freedom third. The component library is substantial — **C3, C4, C5 and C8 cyclic rotors** and **D2, D3, D4, D5 and D8 dihedral axles** — confirmed by cryo-EM at 6.0 Å (C3 rotor, 28 Å internal diameter), 7.9 Å (C4 rotor, 57 Å internal diameter), 4.2 Å (D3 axle) and 7.4 Å (D8 axle) (Courbet 2022). The payoff is the **symmetry-mismatched** assemblies: **D3–C3** and **D3–C5** axle-rotor systems that assemble in vitro and in vivo and, by cryo-EM, populate conformationally variable relative orientations reflecting the symmetry of the coupled components and the designed interface energy landscape — the D3–C3 system largely constrained, the mismatched D3–C5 rotor free to rotate and translate along the main symmetry axis (Courbet 2022).

The result is a genuine conceptual advance: **symmetry as a mechanical variable**, where matching symmetries rigidify and mismatching them leaves a shallow periodic landscape along a rotational coordinate. But the supplement shows what it cost. Reaching a D3 axle required sampling **two different starting superhelical radii per helix (6 Å and 12 Å)** over a ~5 Å range in 0.5 Å bins, with helical phase swept 0°–90° in 10° steps, z-offsets from −1.51 Å to +1.51 Å in 0.1 Å steps, and supercoil phases fixed by hand at 0° for D3 and 30° for D2 — followed by HBNet interface design, RosettaRemodel helical fusion to cyclic partners, a second design round across the fusion, and SymDofMover for the D8 case (Courbet 2022 (suppl.)). Every symmetry group is a separate engineering project.

5. Hallucination — symmetry as an objective (and, first, as an accident)

5.1 The method, and the oligomers nobody asked for

Hallucination as a design method was established by Anishchenko 2021, and the mechanism is worth stating in its original, symmetry-free form, because everything the symmetric version does is a modification of this. Start from a **random amino acid sequence**, whose predicted inter-residue distance and orientation maps are diffuse and featureless. Run **Monte Carlo simulated annealing in sequence space**, at each step substituting a random residue at a random position, re-predicting with **trRosetta**, and accepting by the Metropolis criterion on the change in **Kullback–Leibler divergence** between the predicted distributions and generic length- and separation-dependent background distributions. The objective is pure **contrast** — make the network's prediction sharp — with no target structure specified anywhere; the authors' stated analogy is DeepDream, inverting a network trained to recognize patterns so that it generates them (Anishchenko 2021).

It worked, and the numbers set the baseline against which the symmetric variants should be read. After around **20,000 Monte Carlo steps** the maps were as featured as those for natural sequences; **2,000** proteins were generated; outputs were novel (median BLAST E-value **0.17** to UniRef90, average TM-score **0.78** to the closest PDB match) and spanned all-α, all-β and mixed α/β with **95 sub-folds**. Of **129** designs expressed, **27** gave SEC peaks for monomeric or small oligomeric species with CD spectra consistent with the targets, **21** were

thermostable above 70 °C, and three structures were solved — NMR at **1.82 Å** backbone RMSD (design 0515), X-ray at **2.53 Å** (0217) and X-ray at **3.68 Å** overall but **1.32 Å** and **2.17 Å** across the N- and C-terminal halves (0738_mod) (Anishchenko 2021). A separate, telling detail: the closest PDB structure to 0738_mod is **Top7**, the canonical Rosetta de novo design (Anishchenko 2021 (suppl.)).

And here is why this paper belongs in a chapter about symmetric oligomers. Every sequence was modelled as a **monomer**, and yet **many of the hallucinated proteins formed oligomers in solution**. Design 0217 crystallized as a **dimer**, corroborated by SEC-MALS and NMR rotational correlation time; design 0417 self-associated transiently, which ultimately prevented NMR structure determination (Anishchenko 2021). The diagnosis is specific and, for this chapter, important: **trRosetta was trained on native structures including many homo- and hetero-oligomers, so it never learned to disfavour surface hydrophobic residues to the degree required for soluble monomers** — the network had absorbed the sequence signatures of native protein–protein interfaces, and β-sandwich designs suffered worst, plausibly reflecting immunoglobulin-fold interdomain contacts in the training data (Anishchenko 2021). The rescue was manual and effective: substituting exposed hydrophobics with polar residues converted several oligomers to monomers, demonstrated directly by paired before/after SEC traces (Anishchenko 2021 (suppl.)).

So the first thing a predictor-inverted generator did, unprompted, was **make homo-oligomers** — because interface-forming sequence patterns are what it had learned from the PDB. The wiki's reading is that this reframes what Wicky 2022 achieved a year later. The hard part was never eliciting an oligomer from the network; it was **specifying which one**, converting an uncontrolled propensity into an addressable input.

5.2 Wicky — making valency an input

Wicky 2022 removes the constraint that the protomer be known in advance, and it does so by changing what kind of object symmetry is. Instead of building symmetry into the coordinates, it **rewards** symmetry in a loss function — adding a symmetry term to the confidence objective of Anishchenko 2021 and swapping trRosetta for AlphaFold2.

The method takes only two inputs: a chain length L and an oligomer valency N. Starting from a random sequence, a Monte Carlo search in sequence space feeds N copies of the current sequence into **AlphaFold2** (Jumper 2021) and combines the network's confidence metrics — pLDDT and pTM — with a measure of cyclic symmetry, namely the **standard deviation of the distances between the centres of mass of adjacent protomers** in the predicted structure (Wicky 2022). Trajectories for monomers through heptamers with 65–130 residue chains typically coalesce within a few hundred steps, at a cost of roughly **1 to 7 CPU-days** per design (Wicky 2022). The outputs — called HALs — span all-α, mixed α/β and all-β topologies and differ from the cyclic de novo designs already in the PDB; median TM-scores to the closest PDB relative are **0.67 for protomers and 0.57 for oligomers**, with **29% of structures below the 0.5 fold-assignment cutoff** in CATH/SCOP, indicating substantial generalization beyond the training set (Wicky 2022).

The failure that makes this paper worth reading closely. 150 designs with pLDDT > 0.7 and pTM > 0.7 were selected for experimental testing, and virtually none showed appreciable soluble expression in *E. coli* — median soluble yield **9 mg/L** — with none of the marginally soluble ones showing both the expected SEC oligomerization state and a consistent circular-dichroism profile (Wicky 2022). The authors' diagnosis is **overfitting to the oracle**: hundreds of MCMC steps against AF2 generate adversarial sequences that the network confidently predicts as structured but which have aberrant biophysical properties, analogous to activation maximization producing unrealistic outputs in image classifiers (Wicky 2022).

This is not a new pathology so much as a much more severe instance of a known one. An-

ishchenko 2021 had already reported that hallucinated sequences carry **exposed surface hydrophobics** inherited from the network's training on native oligomeric structures, and had fixed it by hand (Anishchenko 2021 (suppl.)). The wiki's reading is that adding a symmetry term makes the same failure much worse rather than introducing a different one: a cyclic-symmetry objective *rewards* inter-protomer contact, so the optimizer is now being paid to produce exactly the sticky surfaces that hallucination was already prone to. What changed between the two papers is not the diagnosis but the remedy — manual polar substitution on a handful of designs in 2021 (Anishchenko 2021 (suppl.)), versus wholesale sequence replacement by a dedicated inverse-folding network in 2022 (Wicky 2022).

The fix is instructive. Discard the hallucinated *sequences*, keep the hallucinated *backbones*, and redesign the sequences with Dauparas 2022 — 24 to 48 sequences per backbone, validated with AF2. Because dozens rather than hundreds of evaluations are performed, overfitting is far less likely; independently, an updated RoseTTAFold confidently predicted almost all ProteinMPNN sequences while failing on most of the original hallucinated ones. The redesigned set behaved completely differently: of 96 tested, **71 (74%) expressed at high levels** (median yield **247 mg/L**, a ~27-fold increase), **50 (52%) had SEC retention volumes consistent with the target oligomer** (30 of those monodisperse), **at least 21 (22%) had the correct oligomeric state by SEC-MALS**, and **67 of 71 (94%)** had circular-dichroism secondary-structure content consistent with the design. **54 of 71** retained secondary structure to 95 °C.

Structural validation was strong where it was attempted. Crystallization was tried for 19 designs and succeeded for **seven** (three C2, two C3, two C4), all with the correct oligomerization state and a **median Cα RMSD of 0.6 Å** at 1.8–3.4 Å resolution, with side-chain rotamers also matching (Wicky 2022). Six large designs with C6–C42 internal repeat symmetry gave negative-stain 3D reconstructions, and three were solved by cryo-EM at 4.38, 6.51 and 6.32 Å: a C15 homopentamer, a C18 homohexamer and a **C33 homotrimer**, **87, 99 and 100 Å in diameter** and 40–50 Å high, each with a continuous parallel β sheet lining the lumen and outer helices enforcing ring closure, with backbone RMSDs to the design models of **0.81, 1.69 and 2.30 Å** and TM-scores of 0.41–0.49 to anything in the PDB (Wicky 2022). For scale: the highest cyclic symmetry recorded in the PDB for a naturally occurring protein is **C39**, the vault proteins (Wicky 2022). Aggregated over the whole campaign, **27 of 30** designs evaluated by SEC-MALS, nsEM, cryo-EM or crystallography had the intended oligomeric state (Wicky 2022).

What it could not do. Hallucination-based cyclic design **fails for higher-order dihedral, tetrahedral, octahedral and icosahedral symmetries**, probably in part because such structures are far less represented in the PDB and therefore in the oracle's learned prior (Watson 2023). The symmetry objective — a standard deviation over *adjacent* protomer centre-of-mass distances — is intrinsically a statement about a ring. It has no natural generalization to a point group with multiple distinct rotation axes.

6. RFdiffusion — symmetry as a constraint on sampling

Watson 2023 is the pivot of this chapter, and symmetric oligomer design is one of the capabilities it was built to unify — alongside unconditional and topology-constrained monomer design, motif and enzyme active-site scaffolding, and binder design. The model fine-tunes the RoseTTAFold structure-prediction network on a denoising task: starting from noised residue frames, it iteratively denoises to a coherent backbone, with conditioning inputs steering generation. For assemblies, the conditioning input is a **point-group symmetry operator**.

The mechanism is worth stating precisely, because it is the third distinct answer to "how do you make a design symmetric." Given a point group and a monomer chain length, random starting residue frames are generated for a **single** subunit as in the unconditional case, and

$n-1$ copies are placed according to the point group. Because RFdiffusion inherits **equivariance** with respect to rotation and chain relabelling from RoseTTAFold, symmetry is *largely maintained* through the denoising predictions on its own; the model nonetheless **explicitly resymmetrizes at each step**, which changes the structures only slightly. For octahedral and icosahedral architectures only the smallest generating subset of monomers is modelled explicitly — for icosahedra, the subunits at the five-, three- and twofold axes — to reduce compute and memory. Design diversity is tuned by **scaling the noise added at inference**, and an **auxiliary inter- and intra-chain contact potential** guides symmetric sampling; external potentials are the same machinery used for ligand and metal pocket design (Watson 2023 (suppl.)).

The striking result is that **RFdiffusion was not trained on symmetric inputs** and nonetheless achieves high in silico success rates on symmetric oligomers, particularly with the contact potential engaged (Watson 2023). Symmetry here is neither built into a coordinate generator nor rewarded in a loss — it is **projected onto the state at every denoising step**, and the model's equivariance is what makes that projection nearly free.

Coverage. Cyclic designs included topologies not seen in the PDB — two-layer β barrels at **C10**, and mixed α/β topologies at **C8** whose closest TM-aligns in the PDB are **0.47 and 0.43** (Watson 2023). Of **608 designs** selected for experimental characterization, **at least 87** had SEC oligomerization states within the 95% confidence interval of the calibration (**126 within the 99% interval**), and negative-stain EM of a **C3 design with 350-residue subunits (1,050 residues total)** matched the design both per-subunit and in overall architecture, including a distinctive pinwheel shape; C5 and C6 designs of more than 750 residues were also consistent with their models (Watson 2023). A separate family of α/β barrel oligomers effectively extends the TIM barrel: **HE0626**, a C6 with 18 strands and 18 helices, and **HE0675**, a C8 with an inner ring of 16 strands and an outer ring of 16 helices in a 1:1 helix:strand pattern locally very similar to a TIM barrel, both confirmed by nsEM 3D reconstruction (Watson 2023). The framing the authors give this is worth keeping: natural evolution extensively explored variations *within* the eight-strand TIM barrel, while a generative model more readily explores **global changes in barrel curvature** (Watson 2023).

Beyond cyclic. This is where RFdiffusion separates from hallucination. SEC indicated that **38 D2, seven D3 and three D4** designs had the expected molecular weights (four, six and eight chains respectively); D3 and D4 2D class averages and 3D reconstructions matched the design topologies, and cryo-EM of the D4 design **HE0537** recapitulated the ~45° offset between tetrameric subunits (Watson 2023). A **12-chain tetrahedron** (HE0964) gave consistent 2D class averages, and of **48 icosahedra** selected for validation, **HE0902** — a **15 nm**, highly porous assembly — formed homogeneous particles whose 2D class averages and 3D reconstruction closely match the design model, with triangular hubs arrayed around empty C5 axes and outward-facing N and C termini suited to antigen display (Watson 2023).

Two things should be read together here. The **coverage** is unprecedented: C, D, T and I in one framework, from noise, with no per-symmetry engineering of the kind Courbet 2022 (suppl.) documents. The **yield** is sobering: 87 of 608 by SEC, one validated icosahedron out of 48 attempts. Generality and success rate are different quantities, and this chapter's reading is that RFdiffusion's contribution to symmetric design is overwhelmingly the former.

7. Three ways to impose symmetry, compared

The three mechanisms are easy to conflate because they produce superficially similar objects. Stated side by side:

Exact by construction. In parametric design the symmetry is a property of the coordinate generator — the superhelical phases are *set* to {0, 2π/n, …} and no other value is repre-

sentable (Huang 2014 (suppl.)). Symmetry error is identically zero. The price is expressivity: only folds the parameterization can write down, in practice helical bundles and barrels (Huang 2014, Thomson 2014).

Rewarded. In hallucination the symmetry is a **term in a loss** — the standard deviation of adjacent-protomer centre-of-mass distances — added to the pure confidence objective that Anishchenko 2021 established and optimized jointly with it (Wicky 2022). Expressivity is enormous: protomer and assembly are discovered together, so intertwined topologies inaccessible to dock-and-design become reachable, and the outputs generalize well past the training set. Three prices are paid. The optimization is expensive (1–7 CPU-days per design). Optimizing *against* a predictor for hundreds or thousands of steps invites adversarial solutions — the 150-design expression failure of Wicky 2022, foreshadowed by the surface-hydrophobic problem already documented in Anishchenko 2021. And the objective is ring-shaped: a statement about *adjacent* protomers does not extend to point groups with several distinct axes, which is why the approach fails for dihedral and cubic symmetries (Watson 2023).

Projected. In diffusion the symmetry is applied to the **state**, at initialization and again at every denoising step, and the model's equivariance means the denoiser's own predictions nearly respect it already (Watson 2023, Watson 2023 (suppl.)). Symmetry error is driven to near zero without ever entering the objective. Expressivity extends to any point group, including cubic ones, and the asymmetric-unit trick keeps icosahedral design tractable. The price is that success is now dominated by whether the *interface* the model invents is physically real — hence the reliance on an auxiliary contact potential, and the 87-of-608 SEC yield.

The through-line the wiki draws from this: **as symmetry moved from the coordinates into the sampling process, the binding constraint moved from geometry to interface quality.** The parametric era's hard problem was writing down a backbone; the diffusion era's hard problem is that writing down backbones is easy and most of them do not assemble.

8. Sequence design is not a detail

It would be easy to treat sequence design as a downstream step. The symmetric-oligomer literature is the strongest evidence in this wiki that it is not.

Dauparas 2022 predicts sequence autoregressively from backbone geometry alone and explicitly handles cyclic oligomers, nanoparticles and interfaces, including **tied/symmetric positions**. The mechanism enabling tying is the **random-order autoregressive decoding** scheme — a detail stated in the main text but derived only in the supplement (Dauparas 2022 (suppl.)) — and the same tied-decoding machinery underlies symmetric design in Watson 2023 (suppl.). Tying matters for a specific reason: under Cn symmetry, the n copies of a residue position are the same residue by definition, so sequence design must either respect that constraint or be corrected afterwards.

The empirical case is the Wicky rescue described above. The same backbones, sequenced two different ways, gave a median soluble yield of **9 mg/L** versus **247 mg/L**, and went from zero validated designs to 27 of 30 with the intended oligomeric state (Wicky 2022). The generator was never the problem; the sequences were. That result also justifies the pipeline shape adopted everywhere afterward — **generate backbone → design sequence with ProteinMPNN → filter with AlphaFold2** (Watson 2023) — and it is a large part of why Dauparas 2022 shows up as a dependency of essentially every later symmetric-design result.

9. What actually gets validated

Symmetric assemblies are unusually well served by orthogonal experimental methods, and the literature has settled into a fairly consistent ladder. It is worth being explicit about it,

because “success rate” means different things at different rungs.

In silico, success is defined by AlphaFold2 self-consistency: **pAE < 5**, **pLDDT > 90** (relaxed to > 80 for motif cases) and backbone RMSD to the design under ~1.0–1.5 Å; these thresholds, not wet-lab data, define the success rates quoted throughout Watson 2023 (Watson 2023 (suppl.)). Hallucination used a looser in-house bar — pLDDT > 0.75 and RMSD to the original backbone < 1.5 Å for the ProteinMPNN redesigns (Wicky 2022).

Experimentally the rungs are: **soluble expression**, then **SEC** retention volume against a calibration curve (cheap, high-throughput, and the metric behind both the 50/96 of Wicky 2022 and the 87/608 of Watson 2023), then **SEC-MALS** for an absolute mass rather than a hydrodynamic proxy, then **circular dichroism** for secondary-structure content and thermal stability, then **negative-stain EM** for shape and 2D class averages, then **cryo-EM** for a 3D reconstruction, and finally **crystallography** for atomic accuracy. The gap between the rungs is real: in Wicky 2022, 52% passed SEC but only ~22% passed SEC-MALS on the same set.

Two observations follow. First, symmetric assemblies are the one design class where **EM is routinely decisive**, because the assemblies are large enough — the giant HAL rings exceeded 100 kDa, which is what made them well suited to EM characterization (Wicky 2022) — and because a ring’s symmetry is directly legible in a 2D class average. Second, the crystallization rate itself carries information: seven of 19 attempted HALs formed diffracting crystals, which the authors note is considerably higher than is typical for Rosetta de novo designs (Wicky 2022).

10. After RFdiffusion

The symmetric-design capability has propagated rather than plateaued. Ingraham 2023, developed independently and contemporaneously at Generate Biomedicines, frames design as **Bayesian inference under external constraints**, with **symmetry** among the first-class conditioners alongside substructure, shape, semantic class and natural-language prompts; conditioners enter as likelihood or constraint terms on the diffusion posterior, giving one uniform mechanism for imposing symmetry (Ingraham 2023 (suppl.)). That two lineages arrived at conditional diffusion with symmetry conditioning separately is, on this wiki’s reading, evidence that the formulation is natural rather than incidental — the same argument made at the level of the whole field in Introduction — the field at a glance.

Butcher 2025 carries the capability into the all-atom, token-plus-atom regime, with a **dedicated inference loop for symmetric design** — token initializers plus symmetry handling — distinct from the main inference loop, explicitly carrying forward the symmetry capability of Watson 2023 (suppl.) (Butcher 2025 (suppl.)). Given that RFdiffusion3 matches or beats its predecessors at roughly **one-tenth the compute** (Butcher 2025), and that symmetric-oligomer success rates are dominated by yield rather than coverage, cheaper sampling is a directly relevant improvement: the 87-of-608 problem is partly a compute-budget problem.

One further thread connects symmetry to function. Watson 2023 scaffolds **symmetric functional motifs** — C3-symmetric trimers rigidly holding three binding domains, and C4-symmetric histidine square-planar metal sites — noting that the symmetry of an assembly can be matched to the symmetry of a coordination geometry or a viral glycoprotein. Symmetric oligomer design and Motif scaffolding (functional-site design) are not separate problems at the level of the model; they are two conditioning inputs to the same generator.

11. Open questions

- **Why is the yield what it is?** No ingested source diagnoses *why* only 87 of 608 symmetric RFdiffusion designs pass SEC (Watson 2023), or whether failures are dominated by

interface quality, kinetic trapping in the wrong oligomeric state, or solubility. The chapter's two mechanistic failure accounts — surface hydrophobics inherited from training on native oligomers (Anishchenko 2021) and adversarial overfitting to the oracle (Wicky 2022) — both concern *hallucination*, and neither transfers cleanly to a diffusion model that never optimizes against a predictor. Whether diffusion's symmetric failures share a cause with hallucination's is open.

- **Did the interface prior survive the transition?** Anishchenko 2021 attributes unintended oligomerization to the network having learned the sequence signatures of native protein–protein interfaces from the PDB. RFdiffusion is fine-tuned from a structure predictor trained on the same kind of data (Watson 2023), so the same prior is presumably present — but whether it *helps* symmetric design (a useful bias toward realistic interfaces) or *hurts* it (a bias toward native-like rather than designed interfaces) is not addressed by any ingested source.
- **Does success rate decay with symmetry order, and how?** The numbers are suggestive — 38 D2 versus seven D3 versus three D4 (Watson 2023) — but the denominators per symmetry group are not stated in the main text, so no rate can be computed. Watson 2023 (suppl.) indicates per-symmetry SEC success rates are characterized in the SI; extracting them would settle this.
- **Is exact resymmetrization necessary?** Watson 2023 reports that explicit resymmetrization at each step changes structures only slightly, since equivariance largely preserves symmetry anyway. Whether the residual asymmetry is noise or a useful relaxation is not addressed.
- **Can hallucination's expressivity be recovered inside diffusion?** Hallucination discovered intertwined protomer/assembly topologies precisely because it never separated them (Wicky 2022). Whether symmetric diffusion samples that same region, or has quietly reintroduced a bias toward separable protomers, is untested in the ingested sources.
- **Two-component and heteromeric assemblies.** Courbet 2022 achieves hetero-oligomeric coupling by design, and the specificity machinery of Boyken 2016 exists to make orthogonal components possible, but the wiki has no ingested source on designed **two-component** symmetric nanomaterials — see the flagged references below.
- **Benchmarks.** There is no shared benchmark for symmetric-oligomer design comparable to the PPI or motif-scaffolding suites in Butcher 2025 (suppl.), so cross-paper numbers in this chapter are not strictly comparable. This is an instance of the `benchmarks/` gap already noted in Introduction — the field at a glance.

12. Flagged references — the history this chapter cannot yet cite

The sections above are written strictly from ingested sources, per the sourcing discipline in `CLAUDE.md`. Several load-bearing pieces of the historical record are therefore either cited **second-hand through a review** or absent entirely. Supplying these PDFs into `sources/` would let the chapter stand on primary literature. Listed in rough order of how much they would improve the chapter; these are also mirrored in index.

Currently cited only second-hand, through Luo 2016:

1. **Padilla, Colovos & Yeates, *PNAS* 2001 — "Nanohedra: using symmetry to design self-assembling protein cages, layers, crystals and filaments."** Luo 2016 credits this as the founding symmetric-fusion result (the M1-matrix/bromoperoxidase tetrahedral cage). Section 2 of this chapter rests on a review's account of it. This is the single most valuable addition.
2. **The Yeates-lab follow-ups on the crystallographically resolved 12-subunit cage** — the refinement that Luo 2016 describes as producing the homogeneous 16-nm cage

with a 5-nm central opening, again currently reachable only through the review.

Named as prior art by ingested sources, but with no primary paper in `sources/`:

3. **King et al., *Science* 2012 — "Computational design of self-assembling protein nanomaterials with atomic level accuracy."** Appears in the reference list of Luo 2016; the icosahedral-nanocage line it founded is what Courbet 2022 refers to as the fixed-orientation nanomaterials it sets out to move beyond. The chapter currently cannot describe how those cages were designed.
4. **King et al., *Nature* 2014 — "Accurate design of co-assembling multi-component protein nanomaterials."** The two-component branch; needed for the open question in §11.
5. **Bale et al., *Science* 2016 — "Accurate design of megadalton-scale two-component icosahedral protein complexes."** The scale demonstration; also the natural bridge to vaccine-platform applications, which Watson 2023 invokes without this wiki being able to support it.

Closed 2026-08-14: the previous entry here requested Anishchenko et al., *Nature* 2021 as the missing foundation of the hallucination lineage. It has since been supplied and ingested as Anishchenko 2021 (+ Anishchenko 2021 (suppl.)), and §5.1 was rewritten around it. All remaining gaps in this list are in the *pre-deep-learning* history.

Already present in `sources/` but not yet ingested — adjacent branches this chapter deliberately excludes:

6. **`Proteins/Shen_Baker_Science_2018_...De novo design of self-assembling helical protein filaments (+supp).`** Helical rather than point-group symmetry; would support a companion section on unbounded assemblies.
7. **`Proteins/Brunette_Baker_Nature_2015_...Exploring the repeat protein universe through computational protein design.`** Repeat symmetry; relevant because the giant HAL rings have LRR-like protomers (Wicky 2022).
8. **`ProteinDesign/SchneiderDuhovnik_NAR_2005_PatchDock and SymmDock servers for rigid and symmetric docking.`** Would let the "dock-and-design" paragraph in §2 cite a primary symmetric-docking method rather than Wicky 2022's retrospective characterization of the era.

Supplements missing for pages this chapter leans on: Thomson 2014 and Boyken 2016 are both *Science* papers whose supplementary materials contain the actual method descriptions (the barrel scoring function; the HBNet algorithm and scripts). Both are marked `supp: missing-requested`.

13. Sources

Ingested sources feeding this chapter, roughly chronologically:

- Harbury 1993 — oligomeric state is encoded in core residue geometry · *Science* 1993
- Luo 2016 — survey; the symmetric-fusion lineage · *Chem. Rev.* 2016
- Huang 2014 (+ Huang 2014 (suppl.)) — parametric backbones from the Crick equations · *Science* 2014
- Thomson 2014 — designed C5/C6/C7 barrels with defined channels · *Science* 2014
- Boyken 2016 — HBNet; specificity as a separate design axis · *Science* 2016
- Anishchenko 2021 (+ Anishchenko 2021 (suppl.)) — hallucination founded; oligomers as an unintended byproduct · *Nature* 2021
- Dauparas 2022 (+ Dauparas 2022 (suppl.)) — sequence design with tied symmetric positions · *Science* 2022

- Courbet 2022 (+ Courbet 2022 (suppl.)) — symmetry-mismatched machines; the hand-built ceiling · *Science* 2022
- Wicky 2022 — hallucinating cyclic oligomers; giant rings to C33 · *Science* 2022
- Watson 2023 (+ Watson 2023 (suppl.)) — point-group-conditioned diffusion; C, D, T, I · *Nature* 2023
- Ingraham 2023 (+ Ingraham 2023 (suppl.)) — symmetry as a Bayesian conditioner · *Nature* 2023
- Butcher 2025 (+ Butcher 2025 (suppl.)) — dedicated symmetric inference loop, ~10× cheaper · bioRxiv 2025
- Jumper 2021 — the confidence oracle used by both the hallucination objective and the diffusion filter · *Nature* 2021

Related pages: concept Symmetric assembly design · concepts Diffusion generative design, Motif scaffolding (functional-site design), Sequence design (fixed-backbone) · tools RFdiffusion, RFdiffusion3 (RFD3), Chroma, ProteinMPNN, AlphaFold.

Concepts — a lookup section

Binder design

De novo design of proteins that bind a chosen target with high affinity and specificity, from the target structure alone — a route to therapeutics, diagnostics and imaging reagents that bypasses immunization and library screening.

Progression captured so far.

1. **DL-augmented Rosetta** — Bennett 2023: keep the Rosetta/motif pipeline but swap in ProteinMPNN for sequence design and use Jumper 2021 self-consistency as an *in silico* filter → ~10× higher success rate.
2. **Generative (protein targets)** — Watson 2023: diffuse binder backbones directly against target hotspots, then sequence + filter with the same tools.
3. **Ligand-conditioned** — Krishna 2024 (RFdiffusionAA): diffuse a binder backbone *around* a small molecule; experimentally validated for digoxigenin, heme and bilin. Extends de novo binding beyond protein targets to small-molecule ligands.
4. **Flexible/peptide targets** — Vasquez-Torres 2023: binders to floppy α-helical peptide hormones (PTH, glucagon, GLP1, secretin, NPY, BH3) via RFdiffusion + **partial diffusion**, then converted into biosensors. Pushes binder design to conformationally heterogeneous targets.
5. **Nucleic-acid & efficient all-atom** — Butcher 2025: one all-atom model designs protein–protein, protein–DNA and protein–small-molecule interfaces at ~10× less compute.

The discipline. Both rely on "predict-then-order": score fold probability and binding-pose probability before any synthesis, converting a low hit-rate wet-lab search into a filtered shortlist.

Sources: Bennett 2023, Watson 2023, Krishna 2024, Vasquez-Torres 2023, Butcher 2025.

Diffusion generative design

Generating novel protein backbones (and complexes) by learning to reverse a noising process — the paradigm that now dominates de novo structure generation.

Two independent demonstrations (2023). Watson 2023 fine-tunes the RoseTTAFold predictor into a backbone denoiser, excelling at monomers, symmetric assemblies, motif scaffolding and binders. Ingraham 2023 builds a from-scratch diffusion model with a polymer-statistics-aware noise prior and programmable (including natural-language) conditioning. Different lineages, same conclusion: diffusion works for proteins.

Convergence with prediction. Abramson 2024 adopts a diffusion coordinate head for *prediction*, so prediction and design increasingly share machinery.

Going all-atom. Krishna 2024's denoising fine-tune (RFdiffusionAA) diffuses protein backbones *around* a small-molecule ligand — extending diffusion design from protein-only to ligand-conditioned Binder design (validated for digoxigenin, heme, bilin).

Conditioning is the point. The value is steerability — hotspots for binders, symmetry operators, fixed functional motifs, shape/semantic constraints. Backbones are then sequenced by Sequence design (fixed-backbone) (ProteinMPNN) and filtered by Structure prediction.

All-atom generation — three routes. The field is now pushing past backbone-only generation: Krishna 2024 atomizes ligands inside a predictor (RFdiffusionAA); Chu 2024 (Huang lab) makes side-chain generation intrinsic via a "superposition + collapse" scheme, codesigning sequence; and Butcher 2025 uses a token+atom sparse architecture to hit prior quality at ~10× less compute across ligand/nucleic-acid/enzyme tasks.

Open contrast to file: fine-tuned-predictor (RFdiffusion) vs. bespoke correlated-noise prior (Chroma) vs. superposition all-atom (Protpardelle) vs. token+atom sparse (RFD3) — tradeoffs not yet resolved in these sources.

Sources: Watson 2023, Ingraham 2023, Abramson 2024, Krishna 2024, Chu 2024, Butcher 2025.

Motif scaffolding (functional-site design)

Building a full protein *around* a fixed functional site — an enzyme active site, a metal/ligand pocket, a binding or epitope motif — so the site is held in its active geometry. The second half of functional de novo design (the first half being: obtain the motif geometry).

Two generations of method.

1. **Hallucination & inpainting** — Wang 2022 introduced both: *constrained hallucination* (optimize a sequence until the network folds it to contain the motif) and *inpainting/RFjoint* (one forward pass fills sequence+structure around the motif). Flexible vs. fast. The unconstrained hallucination engine both build on comes from Anishchenko 2021, which anticipated this application — noting that because the network hallucinates plausible structures from scratch, no scaffold blueprint need be specified in advance, and the loss can be generalized to include binding motifs or catalytic sites.
2. **Conditional diffusion** — Watson 2023 generalizes these: conditional training scaffolds motifs by denoising (see Watson 2023 (suppl.) §3.2), with higher success and diversity. Butcher 2025 extends to **atomic** motifs (tip-atom conditioning; the AME enzyme benchmark), and Chu 2024 scaffolds motifs **backbone- and rotamer-free** in all-atom space.

Common pattern. Motif in → scaffold out → sequence with ProteinMPNN → filter with AlphaFold2/self-consistency.

Sources: Wang 2022, Watson 2023, Butcher 2025 (and all-atom variant Chu 2024).

Sequence design (fixed-backbone)

Given a target backbone, find an amino-acid sequence that folds to it — the "inverse folding" problem, and the second stage of the dominant de novo recipe.

Shift from physics to learning. Classic approach: Rosetta energy optimization over rotamers — accurate but slow, with limited success on hard targets. Dauparas 2022 replaced it with an autoregressive message-passing network conditioned on backbone geometry, raising native sequence recovery to ~52% (vs. ~33% for Rosetta) in seconds, and generalizing to interfaces, oligomers and nanoparticles. Independently, Anand 2022 (Huang lab) reached the same conclusion from the "learned energy" direction — a network learning the design potential directly from crystal structures with no human priors, generalizing to unseen topologies (incl. a de novo TIM-barrel) with crystal-validated designs.

Toward codesign. The next step folds sequence into structure generation: Chu 2024 codesigns all-atom structure and sequence (miniMPNN in the loop), blurring the line between this concept and Diffusion generative design.

Role in pipelines. Sequence design is the step *after* backbone generation: Watson 2023 (or Ingraham 2023) produces a backbone, then ProteinMPNN assigns sequence, then a predictor filters. Bennett 2023 swapped Rosetta for ProteinMPNN specifically for this efficiency.

Key model: ProteinMPNN.

Sources: Dauparas 2022, Anand 2022, Bennett 2023, Chu 2024.

Structure prediction

Predicting 3D structure from sequence — historically the "protein folding problem." Now largely solved for single chains and increasingly for complexes, and the enabling substrate for modern design.

State of the art. Jumper 2021 reached atomic accuracy at CASP14 using MSA + pair reasoning (Evoformer) and an explicit structure module. Abramson 2024 extends this to protein–ligand, protein–nucleic-acid and antibody–antigen complexes with a diffusion coordinate head. RoseTTAFold (Baker lab) is the parallel lineage whose network is repurposed for generation in Watson 2023; its all-atom extension Krishna 2024 brings small molecules, nucleic acids and metals into a single model — the design-side counterpart to AlphaFold3.

Why it matters for design. Prediction models double as **design oracles**: their confidence scores (pLDDT, PAE, pTM) filter designs *in silico* before synthesis — see Bennett 2023 and the RFdiffusion→MPNN→AF2 loop. Prediction and design are converging on shared diffusion machinery.

Key models: AlphaFold · RoseTTAFold.

Sources: Jumper 2021, Abramson 2024, Krishna 2024, Bennett 2023.

Symmetric assembly design

Designing assemblies in which several copies of one chain are arranged under a **point-group symmetry** — cyclic (C*n*) rings, dihedral (D*n*) stacks, and the cubic groups — as building blocks for nanomachines, biomaterials and multivalent scaffolds.

> **This concept has been promoted to a chapter.** See Symmetric oligomer design for the full treatment: the historical arc, the three mechanisms for imposing symmetry, validation practice, and open questions. This page stays as the short canonical definition to cite from elsewhere.

The one-line definition. Symmetry lets a single designed protomer and a single designed interface specify an assembly of *n* chains, because the relative orientation of adjacent subunits plus the complementarity of their contact interface determines the multiplicity of the complex (Luo 2016).

Three ways to impose it. The distinction matters because the methods are not variations on one idea.

- **Exact by construction** — parametric generation, where symmetry is a property of the coordinate generator: for an *n*-helix bundle the superhelical phase simply takes the values {0, 2π/n, … 2(n−1)π/n} (Huang 2014 (suppl.)). See Huang 2014, Thomson 2014.
- **Rewarded** — deep-network hallucination, where symmetry is a **term in a loss**: the standard deviation of centre-of-mass distances between adjacent protomers in an AlphaFold2 prediction, added to the pure-confidence objective of Anishchenko 2021 and optimized jointly with it, from inputs of only chain length and valency (Wicky 2022).
- **Projected** — conditional diffusion, where point-group operators are applied to the noised state at initialization and the structure is **explicitly resymmetrized at each denoising step**; equivariance means this changes the predictions only slightly (Watson 2023, Watson 2023 (suppl.) §3.1).

Why the third one won on coverage. Hallucination-based cyclic design **fails for higher-order dihedral, tetrahedral, octahedral and icosahedral symmetries**, in part because those are much less represented in the PDB (Watson 2023); diffusion reaches all of them in one framework, using only the smallest generating subset of subunits for the cubic groups. Butcher 2025 carries a dedicated symmetric-design inference loop (Butcher 2025 (suppl.)), and Ingraham 2023 arrives at symmetry conditioning independently, as one Bayesian conditioner among several.

Specificity is a separate axis from geometry. Getting the right shape does not give orthogonal assemblies; buried hydrogen-bond networks enumerated by HBNet confer interaction specificity in vivo and were validated across ten crystal structures (Boyken 2016).

Sequence design is load-bearing, not downstream. The same hallucinated backbones went from a median soluble yield of 9 mg/L to 247 mg/L when the hallucinated sequences were discarded and redesigned with Dauparas 2022 (Wicky 2022); tied/symmetric decoding is what makes this possible (Dauparas 2022 (suppl.)).

Symmetry was a byproduct before it was a target. Unconstrained hallucination, with every sequence modelled as a monomer, nonetheless produced many designs that formed homo-oligomers in solution — design 0217 crystallized as a dimer — because the network had learned the sequence signatures of native protein–protein interfaces and did not disfavour surface hydrophobics enough for soluble monomers (Anishchenko 2021). Substituting exposed hydrophobics with polar residues reversed it (Anishchenko 2021 (suppl.)). Eliciting an oligomer was never the hard part; specifying *which* one was.

Chapter: Symmetric oligomer design · **Sources:** Harbury 1993, Huang 2014, Thomson 2014, Boyken 2016, Luo 2016, Anishchenko 2021, Courbet 2022, Wicky 2022, Watson 2023, Butcher 2025.

Reference

Glossary

Concise definitions of the recurring terms in this book. Where a term has its own chapter or concept entry, follow its cross-references for the full treatment.

AF2 / AF3 AlphaFold 2 and AlphaFold 3, DeepMind's structure predictors. AF2 predicts protein monomers and complexes from an MSA; AF3 replaces the structure module with a diffusion decoder and generalises to nucleic acids, ligands and modified residues.

All-atom design Generative design that places side-chain (and often ligand) atoms jointly with the backbone, rather than generating a backbone first and assigning sequence afterwards. Contrast *two-stage design*.

Backbone The repeating N–Cα–C(=O) chain of a protein, ignoring side chains. Most first-generation generative models produce backbones only.

Binder A designed protein that binds a specified target with useful affinity and specificity. See the binder-design chapter.

Chroma Generate Biomedicines' programmable generative model, which diffuses under a polymer-statistics prior and steers sampling with modular conditioners.

Codesign Generating structure and sequence simultaneously in one model, as opposed to designing sequence onto a fixed backbone.

Conditioner A module that biases a diffusion model's sampling toward a constraint (symmetry, shape, substructure, a natural-language or property target) without retraining the base model.

CASP Critical Assessment of Structure Prediction — the biennial blind community benchmark on which AlphaFold2's accuracy was first established.

Denoising / reverse diffusion The generative process in a diffusion model: start from noise and iteratively remove it, each step conditioned on the model's current estimate of the clean structure.

De novo design Designing a protein from scratch rather than modifying a natural one.

Diffusion model A generative model trained to reverse a gradual noising process; the dominant architecture for structure generation (RFdiffusion, Chroma, Protpardelle, RFdiffusion3).

Evoformer AlphaFold2's trunk: paired sequence and residue-pair representations updated by attention with triangle operations that enforce geometric consistency.

FAPE (frame-aligned point error) AlphaFold2's training loss, comparing predicted and true atom positions in every residue's local frame — rotation- and translation-invariant, and sensitive to local as well as global error.

Foldseek A fast structural-alignment search tool; used in design pipelines to test whether a generated backbone resembles anything in the PDB (i.e. how novel it is).

Fold switching The ability of one sequence to adopt more than one distinct fold — a known failure mode for single-conformation structure prediction.

Hallucination Design by optimising a sequence so that a structure predictor confidently folds it into something — the input is an objective, not a target backbone. The route used for early motif scaffolding and symmetric assemblies.

Hotspot residues Target-surface residues specified as the intended contact patch when designing a binder.

Inpainting Filling in missing structure or sequence around a fixed region, the complement of hallucination for motif scaffolding.

Inverse folding Predicting a sequence that will fold into a given backbone — the problem ProteinMPNN solves. Synonym in practice for *fixed-backbone sequence design*.

IPA (invariant point attention) The attention mechanism in AlphaFold2's structure module that operates on 3D points in residue frames and is invariant to global rotation and translation.

lDDT / pLDDT The local distance difference test, a superposition-free measure of local structural accuracy; **pLDDT** is a model's per-residue self-estimate of it, widely reused as a design filter.

MaSIF Molecular surface interaction fingerprints — geometric deep learning on protein surfaces, used to find and design interaction patches.

Motif scaffolding Building a new protein around a fixed functional site (a binding motif, an active site) so the motif's geometry is preserved.

MPNN Message-passing neural network; in this field usually shorthand for ProteinMPNN.

MSA (multiple sequence alignment) A stack of evolutionarily related sequences whose covariation encodes contact information; the input signal that made AlphaFold2 accurate, and a dependency later models try to reduce.

Oracle A predictor used to score designs *in silico* before synthesis — typically AlphaFold2, which both filters designs and (via hallucination) can be optimised against.

pAE / PAE (predicted aligned error) A predicted pairwise error matrix; `pAE_interaction` across a designed interface is a standard in silico discriminator of binders that will work experimentally.

Partial diffusion Noising an existing design only part-way and re-denoising it, to sample local variants around a promising solution rather than starting from scratch.

PDB The Protein Data Bank — the archive of experimentally determined structures that trains and benchmarks essentially every model in this book.

ProteinMPNN The Baker-lab graph neural network for fixed-backbone sequence design; fast, high sequence-recovery, and the default sequence step in most design pipelines.

Protpardelle The Huang-lab all-atom generative model, which handles side chains via a superposition over side-chain states.

RFAA (RoseTTAFold All-Atom) The Baker-lab generalised predictor covering proteins with small molecules, nucleic acids and covalent modifications; the basis for RFdiffusionAA.

RFdiffusion The Baker-lab diffusion model that generates protein backbones, fine-tuned from RoseTTAFold; the workhorse for de novo backbones, motif scaffolding, symmetric oligomers and binders.

RFdiffusion3 The all-atom successor generating tokens and atoms directly, substantially cheaper to sample and extended to nucleic-acid and small-molecule interactions.

RMSD (root-mean-square deviation) The standard measure of structural agreement between two aligned structures; design success is usually defined by an RMSD threshold between the design model and the predictor's refolded prediction.

Rosetta The physics- and knowledge-based macromolecular modelling suite that defined computational protein design before deep learning, and still supplies energy functions and relax/repack protocols.

Scaffold (1) The designed protein framework that presents a functional motif; (2) a reusable natural framework (nanobody, monobody, fibronectin III domain) engineered for new binding.

Self-conditioning Feeding a diffusion model its own previous-step structure prediction as extra input; a large empirical accuracy gain in RFdiffusion.

Sequence recovery The fraction of native residues a sequence-design model reproduces when given a native backbone — the standard in silico metric for inverse folding.

Structure module AlphaFold2's final block, which converts the Evoformer representations into explicit 3D coordinates via residue frames and IPA.

Symmetric assembly A designed homo-oligomer with cyclic (Cn), dihedral (Dn) or point-group symmetry; symmetry is imposed during generation rather than discovered.

Test-time compute Spending more sampling or search effort per design at inference (more seeds, more denoising steps, re-ranking) to buy success rate without retraining.

Tokenization How a model discretises a biomolecular system into units — residues, atoms, or a mixture — which determines what kinds of molecules it can represent.

Two-stage design The dominant pipeline: generate a backbone (RFdiffusion), then design a sequence onto it (ProteinMPNN), then verify by refolding (AlphaFold2). Contrast *co-design* and *all-atom design*.

Yeast display A high-throughput experimental screen presenting designed proteins on the yeast surface, used to measure binding across large design libraries.

References

Every source ingested into the wiki, with a one-line description. In-text citations throughout the book link here.

Abramson 2024. Abramson et al. (DeepMind / Isomorphic Labs). Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 2024. — Generalizes Jumper 2021 from single proteins to the joint structure of complexes spanning proteins, nucleic acids, small molecules, ions and modified residues, within one unified model.

Abramson 2024 (suppl.). Accurate structure prediction of biomolecular interactions with AlphaFold 3 — Supplementary Information. — Methods/architecture companion to Abramson 2024. 45 pages of pseudocode-level specs. Best read against the Jumper 2021 (suppl.) to see exactly what changed from AF2.

Anand 2022. Anand, Eguchi, Mathews, Perez, Derry, Altman, Huang (Huang lab, Stanford). Protein sequence design with a learned potential. Nature Communications 2022. — A Huang-lab deep-learning model for fixed-backbone Sequence design (fixed-backbone) that learns entirely from crystal-structure data with **no human-specified energy terms** — an early demonstration that learned potentials can replace hand-crafted ones.

Anishchenko 2021. Anishchenko, Pellock, Chidyausiku et al. (Baker lab). De novo protein design by deep network hallucination. Nature 2021. — The paper that **inverted a structure predictor into a generator**. Random sequences are optimized until the network's own predictions become confident, and the resulting sequences fold — establishing hallucination as a design method and founding the lineage that runs through Wang 2022 and Wicky 2022 into conditional diffusion.

Anishchenko 2021 (suppl.). De novo protein design by deep network hallucination — Supplementary Information. — Companion to Anishchenko 2021. A short SI: one table and seven figures. It is unusually load-bearing for its length, because **most of it documents the method's failure modes** rather than its successes — in particular the unintended oligomerization that matters to Symmetric oligomer design.

Bennett 2023. Bennett, Coventry et al. (Baker lab). Improving de novo protein binder design with deep learning. Nature Communications 2023. — Shows that inserting deep-learning models into the classic Rosetta binder pipeline — as sequence designer and as *in silico* filter — raises success rates roughly ten-fold.

Bennett 2023 (suppl.). Improving de novo protein binder design with deep learning — Supplementary Information. — Data/validation companion to Bennett 2023. 19 pages, mostly supplementary figures — the retrospective analysis that establishes *which* AF2 metrics discriminate real binders and *how well*.

Boyken 2016. Boyken, Chen, Groves et al. (Baker lab). De novo design of protein homo-oligomers with modular hydrogen-bond network–mediated specificity. Science 2016. — Introduces **HBNet** and uses it to design cyclic homo-oligomers whose **interaction specificity**

is set by modular arrays of buried hydrogen-bond networks — the protein analogue of Watson–Crick base pairing — rather than by hydrophobic packing alone.

Butcher 2025. Butcher, Krishna, Mitra et al. (Baker lab). De novo Design of All-atom Biomolecular Interactions with RFdiffusion3. bioRxiv 2025 (preprint). — The current frontier of the Baker-lab diffusion line: one all-atom model that designs proteins in the context of ligands, nucleic acids and arbitrary non-protein atoms, at **~10× lower compute** than its predecessors.

Butcher 2025 (suppl.). De novo Design of All-atom Biomolecular Interactions with RFdiffusion3 — Supplementary Information. — Methods/benchmark companion to Butcher 2025. 48 pages. Specifies the training regimen, tokenization, the DiffusionModule, and the four benchmarks behind the efficiency and capability claims.

Chu 2024. Chu, Kim, Cheng, El Nesr, Xu, Shuai, Huang (Huang lab, Stanford). An all-atom protein generative model (Protpardelle). PNAS 2024. — A Huang-lab all-atom diffusion model that generates backbone **and** side chains together, via a novel "superposition" trick for handling the mixed continuous/discrete nature of proteins. An independent, distinct approach to all-atom generation from the Baker-lab line.

Chu 2024 (suppl.). An all-atom protein generative model (Protpardelle) — Supplementary Information. — Theory/architecture companion to Chu 2024. 30 pages (Tables S1–S9). Formalizes the superposition scheme and specifies the networks — the detail needed to understand *why* all-atom codesign works here.

Courbet 2022. Courbet, Hansen, Hsia et al. (Baker lab). Computational design of mechanically coupled axle-rotor protein assemblies. Science 2022. — Builds nanoscale **machines** out of designed cyclic and dihedral symmetric components, deliberately **mismatching the symmetries** of an axle and a rotor so that the coupled system retains an internal degree of freedom rather than locking into one rigid structure.

Courbet 2022 (suppl.). Computational design of mechanically coupled axle-rotor protein assemblies — Supplementary Materials. — Methods companion to Courbet 2022. Figs. S1–S22, Tables S1–S3, Movies S1–S5, Data S1–S5. This page extracts the design protocol for symmetric components, which is the part most reusable outside the paper's specific machines.

Dauparas 2022. Dauparas et al. (Baker lab). Robust deep learning–based protein sequence design using ProteinMPNN. Science 2022. — The workhorse sequence-design model: given a backbone, find an amino-acid sequence that folds to it. Replaced Rosetta as the default for the fixed-backbone Sequence design (fixed-backbone) step in most modern pipelines.

Dauparas 2022 (suppl.). Robust deep learning–based protein sequence design using ProteinMPNN — Supplementary Information. — Methods companion to Dauparas 2022. 25 pages (Figs S1–S12, Table S1). Documents the network internals, the ablation experiments that produced the final design, and training-data construction — with pseudocode for the encoder/decoder layers.

Harbury 1993. Harbury, Zhang, Kim, Alber. A Switch Between Two-, Three-, and Four-Stranded Coiled Coils in GCN4 Leucine Zipper Mutants. Science 1993. — The experiment that showed **oligomeric state is encodable in sequence** — that the number of chains in a symmetric assembly can be dialled by changing which hydrophobic residues sit at the core positions of a heptad repeat.

Huang 2014. Huang, Oberdorfer, Xu et al. (Baker lab). High thermodynamic stability of parametrically designed helical bundles. Science 2014. — Designs protein backbones by **varying the parameters of the Crick coiled-coil equations** rather than by sampling fragments, yielding hyperstable bundles — including a symmetric pentamer — whose crystal structures match the design models with nearly identical core packing.

Huang 2014 (suppl.). High thermodynamic stability of parametrically designed helical bundles — Supplementary Materials. — Methods companion to Huang 2014. This page maps the supplement and extracts the parameterization itself, which the main text states only in summary.

Ingraham 2023. Ingraham et al. (Generate Biomedicines). Illuminating protein space with a programmable generative model. Nature 2023. — A diffusion generative model for proteins and complexes developed independently of the Baker lab (Generate Biomedicines), notable for programmable conditioning — including natural-language and geometric constraints.

Ingraham 2023 (suppl.). Illuminating protein space with a programmable generative model — Supplementary Information. — Theory/methods companion to Ingraham 2023. 123 pages — the mathematical core behind Chroma's distinctive design choices (correlated-noise diffusion, sub-quadratic graph network, programmable conditioners).

Jumper 2021. Jumper et al. (DeepMind). Highly accurate protein structure prediction with AlphaFold. Nature 2021. — The paper that effectively solved single-chain protein structure prediction from sequence, the founding result of the modern deep-learning era in Structure prediction.

Jumper 2021 (suppl.). Highly accurate protein structure prediction with AlphaFold — Supplementary Information. — Methods/architecture companion to Jumper 2021. 62 pages, largely pseudocode-level algorithm specifications — the canonical reference for how AF2 actually works.

Krishna 2024. Krishna, Wang, Ahern et al. (Baker lab). Generalized biomolecular modeling and design with RoseTTAFold All-Atom. Science 2024. — Extends structure prediction and design beyond protein-only systems to full biomolecular assemblies — proteins with nucleic acids, small molecules, metals and covalent modifications — and, via a diffusion variant, designs proteins around small-molecule ligands.

Krishna 2024 (suppl.). Generalized biomolecular modeling and design with RoseTTAFold All-Atom — Supplementary Information. — Methods/benchmark companion to Krishna 2024. 107 pages (Figs S1–S34). This is the clean, text-based counterpart to the OCR-recovered main PDF — the authoritative source for RFAA's featurization, training, benchmarks, and the RFdiffusionAA binder-design protocols.

Luo 2016. Luo, Hou, Bai, Wang, Liu. Protein Assembly: Versatile Approaches to Construct Highly Ordered Nanostructures. Chemical Reviews 2016. — A 60-page *Chemical Reviews* survey of how ordered protein nanostructures are built. Ingested for the wiki mainly as a **historical reference for the pre-deep-learning symmetry literature**, which the rest of `sources/` does not otherwise cover.

Thomson 2014. Thomson, Wood, Burton et al. (Woolfson lab). Computational design of water-soluble α-helical barrels. Science 2014. — Designs **cyclic homo-oligomers of five, six and seven parallel identical helices** with open central channels, by combining coiled-coil geometry, knowledge-based scoring and atomistic modeling — the Woolfson-lab counterpart to the Baker-lab parametric work published in the same issue.

Vasquez-Torres 2023. Vázquez Torres, Leung, Venkatesh et al. (Baker lab). De novo design of high-affinity binders of bioactive helical peptides. Nature 2023. — Applies the generative design stack to a hard, high-value target class: **flexible α-helical peptide hormones** (parathyroid hormone, glucagon, secretin, neuropeptide Y, GLP1, GIP, BH3 peptides), producing high-affinity binders and turning them into biosensors.

Vasquez-Torres 2023 (suppl.). De novo design of high-affinity binders of bioactive helical peptides — Supplementary Information. — Methods companion to Vasquez-Torres 2023. 32 pages, heavily **experimental** — the wet-lab protocols behind the designs, which the main text compresses.

Wang 2022. Wang, Lisanza, Juergens et al. (Baker lab). Scaffolding protein functional sites using deep learning. Science 2022. — Introduces the two deep-learning routes to **motif scaffolding** — building a protein around a fixed functional site — that RFdiffusion later generalized. The conceptual bridge from prediction networks to functional de novo design.

Watson 2023. Watson, Juergens, Bennett et al. (Baker lab). De novo design of protein structure and function with RFdiffusion. Nature 2023. — The first broadly successful diffusion generative model for protein backbones, and the general-purpose engine behind much current de novo design.

Watson 2023 (suppl.). De novo design of protein structure and function with RFdiffusion — Supplementary Information. — Methods/data companion to Watson 2023. 129 pages. This page maps the SI and extracts the concrete details the main summary omits — architecture, diffusion math, training/conditioning recipes, and the *in silico* success criteria that define "success" throughout the paper.

Wicky 2022. Wicky, Milles, Courbet et al. (Baker lab). Hallucinating symmetric protein assemblies. Science 2022. — Uses deep-network **hallucination** to design symmetric homo-oligomers from almost nothing — just the number of protomers and their length — producing assemblies unlike anything in the PDB.

AlphaFold. DeepMind's protein structure predictor; the field-defining model and, in practice, the standard design oracle.

Chroma. Programmable diffusion generative model for proteins and complexes from Generate Biomedicines. Introduced in Ingraham 2023.

ProteinMPNN. Baker-lab message-passing network for fixed-backbone Sequence design (fixed-backbone). Introduced in Dauparas 2022.

Protpardelle. Huang-lab all-atom protein diffusion model that generates backbone and side chains together. Introduced in Chu 2024.

RFdiffusion. Baker-lab diffusion generative model for protein backbones, made by fine-tuning the RoseTTAFold predictor on a structure-denoising task. Introduced in Watson 2023.

RFdiffusion3 (RFD3). Baker-lab all-atom diffusion generator for proteins in the context of ligands, nucleic acids and metals — the efficiency-focused successor to RFdiffusion and RFdiffusionAA. Introduced in Butcher 2025.

RoseTTAFold. Baker-lab protein structure predictor — the parallel lineage to DeepMind's AlphaFold, and the backbone repurposed for generative design.